ProFeatX: a parallelized protein feature extraction suite for machine learning

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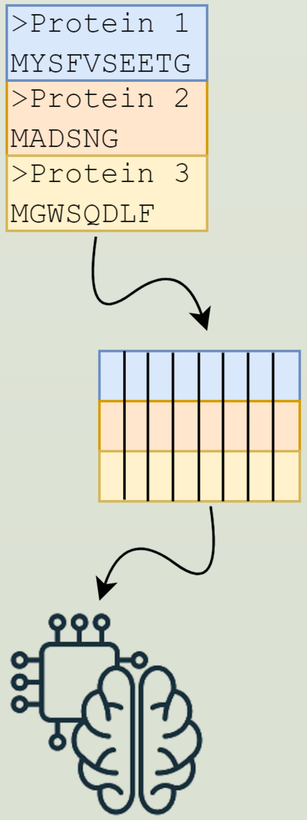
INTRODUCTION

Machine Learning (ML) algorithms have been successfully applied in proteomics, genomics, transcriptomics and have helped the biological community to answer several complex questions. However, most machine learning methods require lots of data, with every data point having the same vector size. The biological sequence data such as proteins are amino acid sequences of variable length, which makes it essential to extract a definite number of features from all the proteins for them to be used as input into ML.

There are numerous methods to achieve this, but only several tools let researchers encode their proteins using multiple methods without having to use different programs or, in many cases, code these algorithms themselves, or even come up with new algorithms. ProFeatX is a tool that contains 50 diverse encodings to extract protein features in an efficient and fast way supporting desktop as well as high-performance computing environment in its standalone version, while having an easy-to-use web interface allowing non-experts to use feature extraction techniques.



METHODS

- ProfeatX was implemented in C++, plus OpenMP for parallelization.
- Web server implemented with Django
- DNA/RNA => Amino acid sequences with TransDecoder

RESULTS

STANDALONE VERSION
- 50 total descriptors
- Parallelization

WEB VERSION
- Runs standalone version on the background with 8 threads
- 31 total descriptors
- Accepts DNA/RNA sequences
- PPI prediction
- Available at http://bioinfo.usu.edu/profeatx/

WEB VERSION DESCRIPTORS

Descriptor	Vector size
AAC	20
DPC	400
TPC	8000
CKSAAP	2400
DDE	400
EAAC	(sequence length – 6) * 20
AAPAS	400
CMV	20
GAAC	5
EGAAC	(sequence length – 4) * 5
CKSAAGP	150
GDPC	25
GTPC	125
EBGW	60
QSO	100
SOCN	60
Geary	240
Moran	240
NMB	240
CTDC	42
CTDT	42
CTDD	210
Conjoint Triad	343
KSCT	646
PseAAC	50
APseAAC	80
Binary	sequence length * 20
TVD	sequence length
PseKRAAC	4
AAIndex	sequence length * 531
BLOSUM62	sequence length * 20
ZScale	sequence length * 5

TIMING BENCHMARKS

Tool	Total descriptors	Time (ms)
ProFeatX	50	764
iFeature	38	4,560
iLearnPlus	37	5,102
MathFeature	14	159,653
Pfeature	37	26,799
protr	25	4,963
Rcpi	22	4,943

These tests were performed for a FASTA file with 22.487 sequences and AAC encoding. ProFeatX was executed on a single thread.

FUTURE RESEARCH

- Increase the number of descriptors.
- Add the PPI and DNA/RNA functionalities to the standalone version.
- Add a feature (dimensionality) reduction functionality.

FUTURE RESEARCH

http://bioinfo.usu.edu
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